

AI MENTAL HEALTH

ANA – 1 (WEEK-3)

IAPT Referral-To-Treatment Analysis

Dataset Used: Psychological Therapies Annual Report (IAPT Dataset)

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1. Introduction

Improving access to psychological therapies is a major NHS objective under the Improving Access to Psychological Therapies (IAPT) programme. Referral-to-Treatment (RTT) waiting time is an important performance indicator used to measure service accessibility and operational efficiency.

This project analyses RTT waiting times using the IAPT annual report dataset. The analysis focuses on identifying variations in waiting times across NHS organizations/regions and disorder categories.

2. Objectives

The objectives of this analysis were:

- To calculate mean RTT waiting times
- To calculate median RTT waiting times
- To compare RTT waiting times across NHS organizations
- To analyze RTT waiting times by disorder type
- To visualize RTT patterns using charts
- To export analytical results for reporting and dashboard development

3. Dataset Description

Dataset Used:

- Psychological Therapies Annual Report (IAPT Dataset)

Dataset Characteristics:

- Contains psychological therapy service statistics
- Includes waiting time indicators
- Includes organization-level and condition-level records
- Includes RTT measures for:
 - Accessing services
 - Completing treatment

Key Variables Used:

Variable	Description
Mean_WaitAccessingServices	Average waiting time to access services
Mean_WaitFinishedCourseTreatment	Average waiting time to complete treatment
OrgName	NHS organization/provider name
VariableType	Category classification
VariableA	Disorder/condition label

4. Data Preprocessing

Several preprocessing steps were performed before analysis.

4.1 Data Cleaning

- Missing values were identified
- RTT columns were converted into numeric format
- Non-numeric symbols such as * and commas were removed
- Rows with completely missing RTT values were excluded

4.2 Data Validation

The dataset structure and variable names were checked to ensure:

- Correct RTT fields were selected
- Organization names were valid
- Disorder categories were available

4.3 Filtering

Separate datasets were created for:

- Regional RTT analysis
- Disorder-type RTT analysis

5. Methodology

The analysis used descriptive statistical techniques.

5.1 Mean RTT Calculation

Mean RTT was calculated using:

$$\text{Mean RTT} = \frac{\sum x_i}{n}$$

where:

- x_i represents RTT values
- n represents total observations

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Mean RTT for Accessing Services: 39.68
Mean RTT for Finished Treatment: 52.17
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5.2 Median RTT Calculation

Median RTT represents the middle RTT value after sorting observations.

Median was used because:

- RTT distributions can contain outliers
- Median provides robust central tendency estimation

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Median RTT for Accessing Services: 31.6
Median RTT for Finished Treatment: 40.25
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5.3 Group-wise Analysis

Data was grouped by:

- NHS organization
- Disorder category

Aggregate RTT metrics were then computed.

	OrgName	Mean_RTT_Access	\
2	42ND STREET	184.80	
193	WE ARE SURVIVORS (SALFORD)	185.10	
157	SIGN HEALTH (HQ)	76.60	
70	LGBT FOUNDATION	143.90	
51	HEREFORDSHIRE AND WORCESTERSHIRE HEALTH AND CA...	57.45	
110	NHS HEREFORDSHIRE AND WORCESTERSHIRE INTEGRATE...	57.40	
104	NHS EAST CHESHIRE TALKING THERAPIES	61.70	
63	KINGSTON TALKING THERAPIES	50.25	
182	VITA HEALTH GROUP: NEWCASTLE IAPT	29.15	
40	EVERYTURN SERVICES LTD TALKING THERAPIES (PETE...	48.05	

	Median_RTT_Access	Mean_RTT_Treatment	Median_RTT_Treatment
2	184.80	236.233333	244.40
193	185.10	231.060000	355.10
157	76.60	161.320000	212.20
70	143.90	140.550000	140.55
51	57.45	121.660000	72.10
110	57.40	120.700000	71.80
104	61.70	113.620000	118.90
63	50.25	105.433333	100.25
182	29.15	102.420000	116.30
40	48.05	96.940000	123.90

6. Region-wise RTT Analysis

Regional RTT analysis examined treatment waiting times across NHS providers.

Findings

- Significant variation exists between NHS organizations
- Some providers show substantially longer treatment delays
- High-demand organizations tend to have elevated RTT values

- Median RTT values were often lower than mean RTT values, suggesting skewed distributions and outliers

Interpretation

The variation may result from:

- Workforce shortages
- Regional service demand differences
- Capacity limitations
- Variations in referral volume

7. Disorder-type RTT Analysis

RTT waiting times were analyzed across disorder categories.

Findings

- Complex and long-term mental health conditions generally exhibited higher RTT values
- Common anxiety and depression pathways tended to have lower waiting times
- Some disorder categories showed substantial RTT variability

Interpretation

Possible causes include:

- Complexity of treatment pathways
- Availability of specialist therapists
- Clinical prioritization processes
- Treatment duration differences

8. Data Visualization

The following visualizations were created:

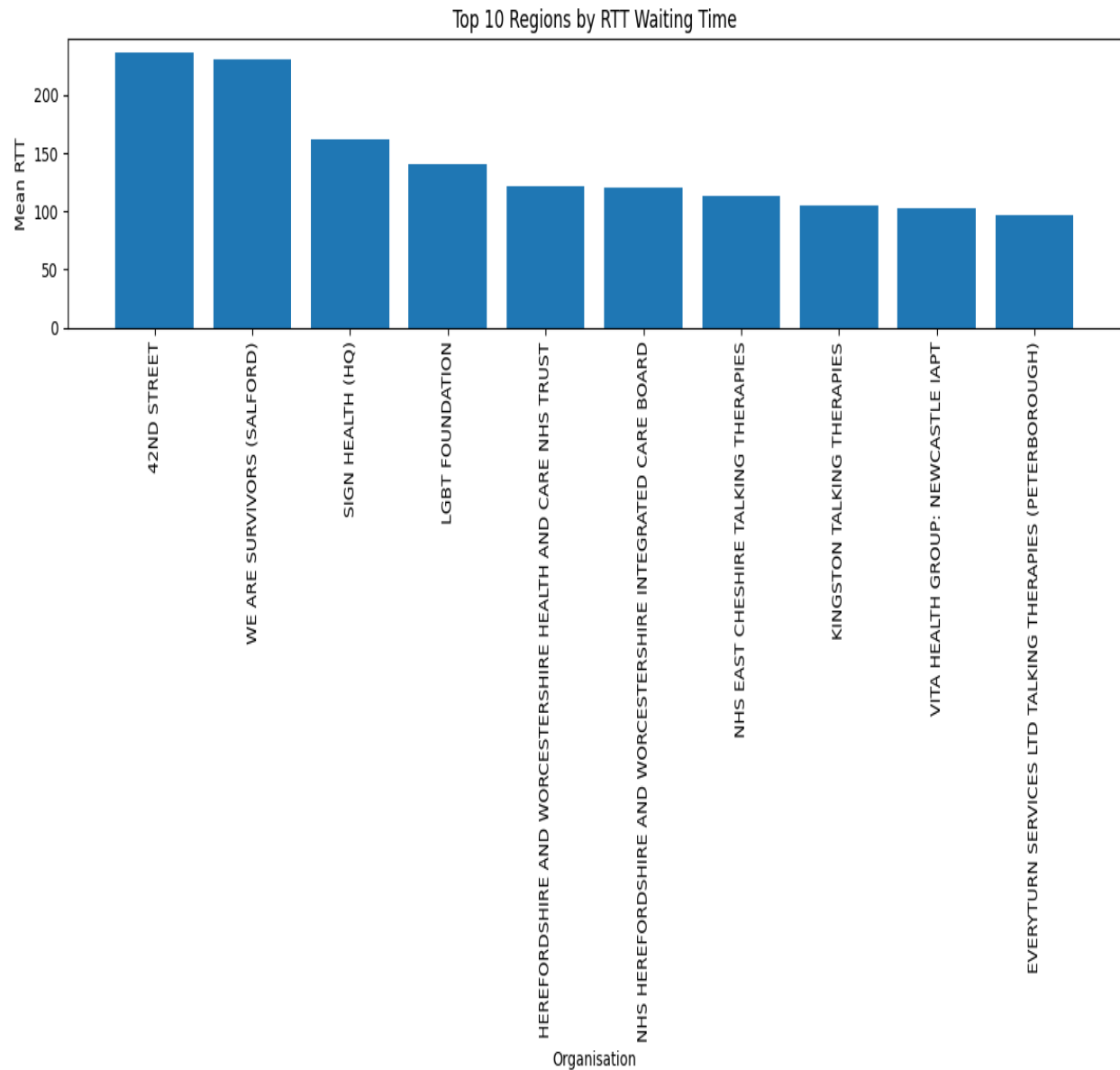
8.1 Region-wise RTT Bar Chart

Purpose:

- Compare RTT waiting times across organizations

Observation:

- A small number of organizations displayed considerably higher waiting times than others



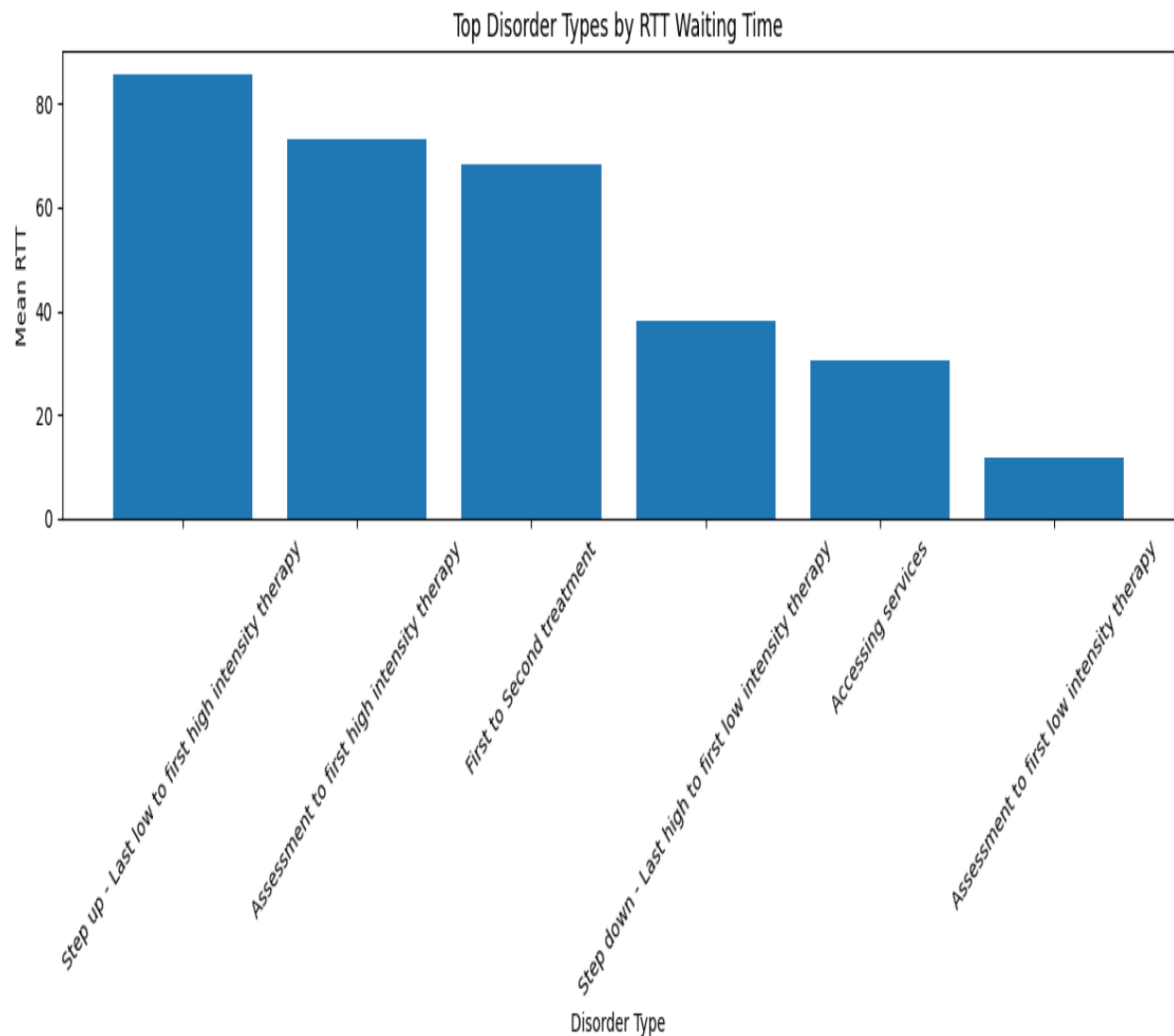
8.2 Disorder-type RTT Bar Chart

Purpose:

- Compare RTT across disorder groups

Observation:

- Certain disorder categories showed disproportionately high RTT values



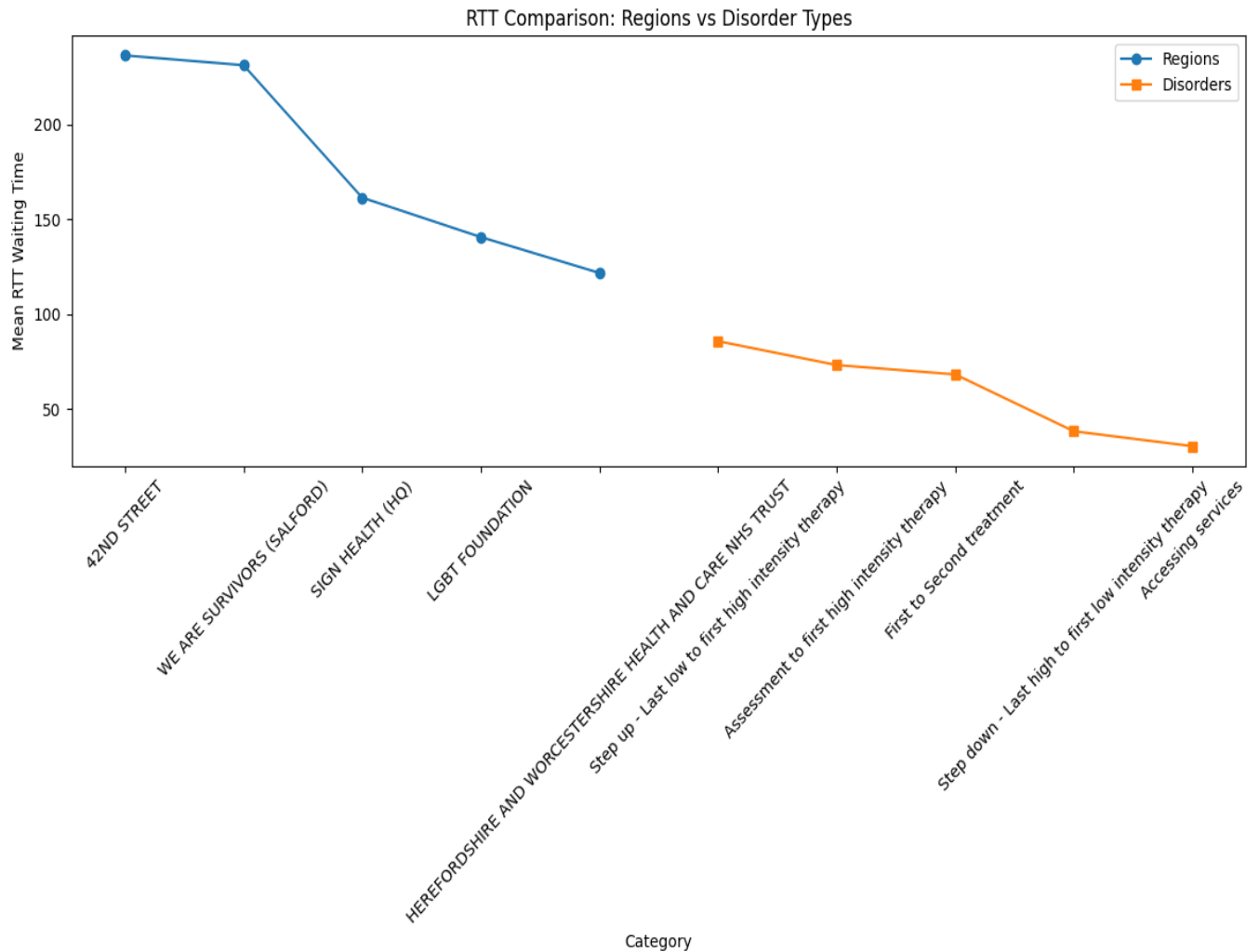
8.3 Combined RTT Comparison Chart

Purpose:

- Compare regional RTT trends with disorder-specific RTT trends

Observation:

- Both organizational and disorder-related factors contribute to RTT variation



9. CSV Export Outputs

The following CSV outputs were generated:

File Name	Description
Region_RTT_Analysis.csv	Region-level RTT statistics
Disorder_RTT_Analysis.csv	Disorder-level RTT statistics

These files can be used for:

- Power BI dashboards
- NHS benchmarking
- Further statistical analysis

10. Conclusion

The IAPT RTT analysis identified meaningful variations in waiting times across NHS providers and disorder categories.

Key conclusions include:

- RTT waiting times are not uniform across organizations
- Complex disorders generally experience longer waiting periods
- Some providers may require additional workforce or capacity support
- RTT distributions contain outliers that significantly influence averages

The analysis provides useful insights for:

- NHS operational planning
- Resource allocation
- Service optimization
- Mental health policy evaluation